WORKSHEET 6.12

Integrals from Graphs and Tables

Unit 6: Integration and Accumulation of Change

Key ideas

- From a graph, a definite integral is a sum of **signed areas** of triangles, rectangles and parts of circles.
- From a table, an integral can only be **approximated**, with a Riemann or trapezoidal sum.
- But $\int_a^b f'(x)\,dx = f(b) - f(a)$ is **exact**, from two values of f.
- The properties of integrals apply as usual: constants, sums and reversed limits.

Worked examples

Example 1 · Areas

The graph of g is the line segment from $(0, 3)$ to $(6, -3)$. Evaluate $\int_0^6 g(x)\,dx$ and $\int_0^6 |g(x)|\,dx$.

Solution. Triangles of area $\frac{9}{2}$ above the axis and $\frac{9}{2}$ below.

0 **and** 9**.**

Example 2 · A table

Selected values of h are shown.

x	0	2	6	7
$h(x)$	5	9	3	1

Approximate $\int_0^7 h(x)\,dx$ with a trapezoidal sum.

Solution. $\frac{1}{2}(5+9)(2) + \frac{1}{2}(9+3)(4) + \frac{1}{2}(3+1)(1) = 14 + 24 + 2$.

40**.**

Example 3 · The integral of a derivative

For the function h above, assumed differentiable, find $\int_0^7 h'(x)\,dx$.

Solution. $h(7) - h(0)$.

-4**, exactly.**

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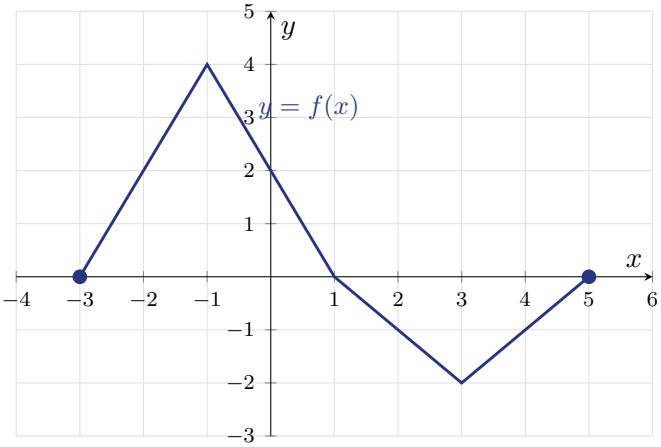
Practice

Questions 1 to 7 use this graph of f, which consists of four line segments.



CORE

1. Find $\int_{-3}^{1} f(x)\,dx$.

2. Find $\int_{1}^{5} f(x)\,dx$.

3. Find $\int_{-3}^{5} f(x)\,dx$.

4. Find $\int_{-3}^{5} |f(x)|\,dx$.

EXAM LEVEL

5. Find $\int_{-1}^{3} f(x)\,dx$.

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6. Find $\int_{-3}^{1} [2f(x) + 1]\, dx$.

7. Find $\int_{-1}^{3} f'(x)\, dx$.

8. A differentiable function k has $k(2) = 11$ and $\int_{2}^{9} k'(x)\, dx = -6$. Find $k(9)$.

Question 9 · Free-response practice

The function h is differentiable on $[0, 10]$. Selected values are shown.

t	0	4	6	10
$h(t)$	8	14	11	3

(a) Approximate $\int_{0}^{10} h(t)\, dt$ with a trapezoidal sum using the three subintervals of the table.

(b) Find the exact value of $\int_{0}^{10} h'(t)\, dt$.

(c) Find the exact value of $\int_{4}^{10} [h'(t) + 2]\, dt$.

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Worked solutions

1. A triangle of base 4 and height 4: 8.
2. A triangle of base 4 and depth 2: -4.
3. $8 - 4 = 4$.
4. $8 + 4 = 12$.
5. On $[-1, 1]$: $\frac{1}{2}(2)(4) = 4$. On $[1, 3]$: $-\frac{1}{2}(2)(2) = -2$. Value 2.
6. $2(8) + 1(4) = 20$.
7. $f(3) - f(-1) = -2 - 4 = -6$.
8. $11 + (-6) = 5$.
9. (a) $\frac{1}{2}(8 + 14)(4) + \frac{1}{2}(14 + 11)(2) + \frac{1}{2}(11 + 3)(4) = 44 + 25 + 28 = 97$.
   (b) $h(10) - h(0) = -5$.
   (c) $(h(10) - h(4)) + 2(6) = -11 + 12 = 1$.

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